

Attention Mechanisms in Sparse Retrieval: A Survey

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Abstract

We survey recent advances in attention mechanisms applied to sparse retrieval. Our analysis covers 78 papers published between 2019 and 2024 and identifies three families of approaches. We further propose a unified evaluation protocol.

Keywords: attention, sparse retrieval, survey, information retrieval

1. Introduction

Attention mechanisms have transformed the field of information retrieval since their introduction. In this paper we systematically review the literature and categorise existing approaches by their architectural assumptions.

2. Methodology

We followed the PRISMA guidelines for systematic reviews. Papers were selected from ACL, SIGIR, NeurIPS and ICLR proceedings using the query terms described in Appendix A. Two annotators independently screened each abstract.

3. Results

Table 1 summarises the surveyed approaches. The majority of methods (61%) rely on learned sparse representations, while 24% use hybrid dense-sparse.

6. Conclusion

We presented a comprehensive survey of attention in sparse retrieval.

References

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